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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 20:45 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.290 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.99 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The accelerated path executes embeddings, decoder layers, final normalization and tied-output GEMV without delegating accelerated tokens to the parent. Wide memory transport and independently owned weight buffers reduce transfer overhead; graph-specific fusion avoids architectural scratch writebacks. Parallel arithmetic increases storage and multiplier resources. A separate literal parent instance preserves legacy modes. The diagram represents interacting responsibilities, not a strictly serial execution sequence.

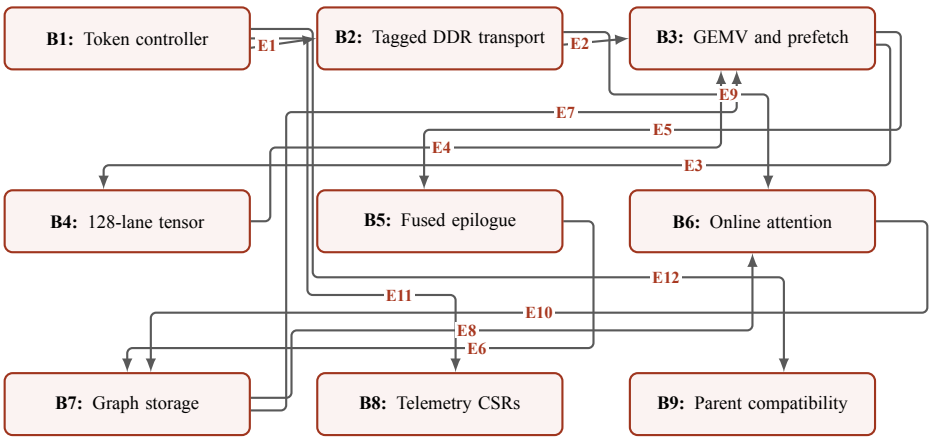


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token controller	Sequences embedding, attention and MLP phases; validates token position/phase; emits ordered logits and manages cancellation.
B2: Tagged DDR transport	Accepts logical 512-bit transactions, retains outstanding ownership and associates reordered responses with descriptors.
B3: GEMV and prefetch	Fetches payloads and scales, assembles two independently owned weight tiles and accumulates row partials in column order.
B4: 128-lane tensor	Uses exact group exponent bases and lane residuals, independently rounded products and seven balanced FP32 reduction levels.
B5: Fused epilogue	Provides RMSNorm replay, two-stage inverse square root, split-half RoPE, SwiGLU, residual addition and pass-through.
B6: Online attention	Broadcasts each causal KV vector across six distinct query states and retains running maxima, denominators and output accumulators.
B7: Graph storage	Holds residuals, BF16 operands, Q/K/V and necessary gate/RoPE branch joins using bounded, phase-owned storage.
B8: Telemetry CSRs	Counts actual activity, stalls, transfers, reuse and completion events; supports atomic snapshots and saturating counters.
B9: Parent compatibility	Instantiates unchanged anvil_q2_fp_top for legal legacy BF16/FP16, AWQ and int8-cache configurations.

Directed relationship	Meaning
E1: B1 to B2	Memory descriptors and cancellation control
E2: B2 to B3	Returned weights and scale metadata
E3: B3 to B4	Tagged weight and activation tiles
E4: B4 to B3	FP32 tile partial sums
E5: B3 to B5	Completed GEMV rows through controller
E6: B5 to B7	Normalized, rotated and joined values
E7: B7 to B3	Registered activation tiles
E8: B7 to B6	Six-head query command
E9: B2 to B6	Controller-assembled causal KV stream
E10: B6 to B7	Normalized attention outputs
E11: B1 to B8	Observed activity and completion events
E12: B1 to B9	Elaboration-selected alternative token path

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, cancel	Input	1 bit each	Single synchronous interface domain; asynchronous active-low transaction reset. Cancellation aborts local work but preserves accepted memory drain obligations.
Token and logit streams	Input/output	32-bit token_ id, token_pos, logit_index and logit_ data; 1-bit controls	Valid/ready handshakes; new_sequence restarts at position zero; token_phase selects prefill/decode. Logits are binary32 in vocabulary-index order. done follows final logit acceptance.
Parent mem_req_* / mem_rsp_*	Output requests/input responses	SB banks; 32-bit data/ coordinates, 64-bit tags	Preserves parent ports and meanings. Kind is 4 bits and projection is 3 bits per bank. Inactive memory interface request outputs are zero.
ddr_req_*	Output, with input ready	512-bit data; 64-bit byte strobes/tag; 32-bit layer/ row/column; 4-bit kind; 3-bit projection; 2-bit format	Logical controller boundary, not a DDR PHY. Every accepted read or write transfers one 64-byte beat and requires exactly one tagged response.
ddr_rsp_*	Input, with output ready	512-bit data, 64-bit tag, valid/error	Responses may reorder. At least two reads can remain outstanding, including QUEUE_DEPTH=1. Live tags cannot be reused; unknown or duplicate tags are faults.
CSR request/response	Input requests/ output responses	16-bit byte address, 32-bit data, 4-bit write strobes	One outstanding request; two registered response stages. Misaligned, unsupported and read-only writes return error. Accessible while token execution stalls.
Tensor streams	Internal input/ output	512-bit weight beats; 2048-bit activations; 128-bit mask; 32-bit tags/ results	Four BF16 or two MX beats assemble a 128-element tile. Each MX beat carries two scale bytes through w_ scale. Weight and activation tiles pair in order with equal tags.
Epilogue stream	Internal input/ output	32-bit operands, coefficients, index, length and tag; 3-bit mode	Modes 0–4 are RMSNorm, RoPE pair, SwiGLU, residual and pass-through. Other modes fail. RMS vectors cannot interleave; stalled payload and metadata remain stable.

Signal / group	Direction	Width	Contract
Attention streams	Internal input/ output	6*HEAD_ DIM BF16 queries; HEAD_DIM BF16 KV vectors; six FP32 output vectors	MX uses the low HEAD_DIM payload bytes and ceil(HEAD_DIM/32) scales. Command precedes KV items; indices must enumerate exactly 0 through cmd_ pos with cmd_len=cmd_pos+1.

Parameter	Default	Meaning / supported values
NUM_LAYERS	28	Accelerated range 1–28.
HIDDEN_SIZE / INTERMEDIATE_SIZE	1536 / 8960	Positive, at most defaults; multiples of eight; inherited divisibility/address checks apply.
NUM_Q_HEADS / NUM_KV_HEADS / HEAD_DIM	12 / 2 / 128	Q heads=6*KV heads; hidden=Q heads*HEAD_DIM; even HEAD_DIM=2–128; KV-head element product divisible by eight.
VOCAB_SIZE / MAX_ SEQ_LEN	151936 / 32768	Accelerated ranges 1–151936 and 1–32768.
GROUP_SIZE / AWQ_ BITS	128 / 4	GROUP_SIZE divides hidden/intermediate; AWQ_BITS= 4. MX grouping is independently fixed at 32.
BANKS / SB	4 / 4	BANKS=1,2,4; SB equals the safe derived bank width.
CACHE_BLOCK / QUEUE_DEPTH	16 / 3	CACHE_BLOCK is a power of two up to 65536 dividing HEAD_DIM; QUEUE_DEPTH=1–16.
FP_FORMAT / WEIGHT_MODE / CACHE_MODE	0 / 0 / 0	Accelerated BF16 uses format 0 and weight/cache modes 0=native or 2=MXFP8. Legal legacy mode-1 selections retain parent semantics.
ACCEL_ENABLE / COUNTER_BITS	1 / 64	ACCEL_ENABLE=0 selects parent; mode 2 requires acceleration and cannot mix with legacy modes. Counter width=1–64.
FlashAttention TILE_ TOKENS	8	Block-level range 1–32, including non-powers of two.

III Behavior and corner cases

A Arithmetic and compression

The numerical contract is binary32 round-to-nearest-even with gradual underflow, signed-zero distinctions and canonical NaN 0x7fc00000. Architectural nonfinite inputs/results fail. BF16 expansion is exact; narrowing is RNE. Group alignment retains full significands and exponent residuals rather than truncating to a common mantissa. MX uses E4M3FN payloads and E8M0 power-of-two scales; scale 255 and NaN payloads are invalid. Encoding selects the smallest scale accommodating the group maximum at payload magnitude 448; zero groups use scale 127.

B Reduction and fusion

Products round independently before the fixed adjacent-pair tree. GEMV tile sums accumulate in ascending column order. RMS squares and sums in index order, multiplies by rounded 1/N, adds epsilon 0x358637bd and applies inverse-square-root normalization. RoPE uses split-half pairs and theta=1000000. SwiGLU evaluates a stable sign-dependent sigmoid using exp(-abs(gate)); exponential arguments below -80 may produce positive zero. Only graph-required residual, RMS replay and branch-join values are retained.

C Pipeline timing

With acceptance at E0, an L-stage pipeline produces valid after E(L-1). Multiply/add/tensor latencies are 3/4/38 registered stages; inverse square root is two stages with initiation interval one. RoPE/residual/pass-through/SwiGLU use 7/4/1/24 stages. RMS inputs are spaced four clocks; its first output is valid 21 clocks after final input acceptance, without stalls.

D Attention

Each KV vector is consumed once and broadcast internally to six independent online-softmax states. Running-maximum changes rescale prior denominators and weighted sums. Final normalization follows the last causal item. Storage does not grow as a sequence-length score matrix. Duplicate indices, malformed last markers and noncausal commands fail.

E Reset, stalls and cancellation

Transfers use pre-edge valid&&ready. Public valid, ready, completion, write and payload outputs are zero during reset. Unreset memories and coefficient captures are protected by valid/ownership state and write-before-read. Stalled items retain complete payloads. Cancel wins over new token acceptance, suppresses new work and waits for accepted tagged requests to drain before cancel_done.

F Telemetry

Version 0x00020200 is at 0x0000; control at 0x0008 provides write-one clear and snapshot. Twelve 64-bit counter pairs span 0x0010–0x0068 for activity, stalls, underflow, bubbles, KV bytes, DDR beats, reuse and completions. Counters saturate and zero-extend. Clear dominates increments; simultaneous clear/snapshot captures zero. Recommendations do not change numerical modes.

IV Verification methods and measured results

A Behavior and compatibility

Method. Compact Icarus checks plus compiled Verilator matrices and independent scoreboards.

Result. Recorded suite passed: 12 legal startup configurations, 29 explicit parameter rejections, seeds 17/89/203, native/MX combinations, legacy compatibility, cancellation/reordering, compact 28-layer sequencing and full-sized operator exercises. This execution predates the nonlinear repair.

Evidence: E087

B Software-vs-Hardware numerical comparison

Method. Simulated RTL outputs compared with independent rational IEEE arithmetic and high-precision mathematics.

Result. Historical snapshot passed 65,536 BF16 patterns, 65,536 MX code/scale pairs, 10,134 finite balanced-tree comparisons, 27,940 positive rsqrt checks within four ULP, 1,000 RMS vectors, 10,000 RoPE pairs, 10,000 SwiGLU elements and 100 six-head attention transactions. Current nonlinear arithmetic has no post-repair numerical execution; no physical FPGA numerical comparison is supplied.

Evidence: E088, E1010, E042

C Defect sensitivity

Method. Disposable, successfully compiled source mutations exercised through genuine comparisons.

Result. All ten defects detected: lane omission, exponent alignment, truncation, buffer association, rsqrt coefficient, RoPE mapping, sigmoid sign, softmax rescaling, sixth-head mapping and telemetry stall omission. Results apply to the recorded snapshot.

Evidence: E089

D Five-domain measured performance

Method. Matched compiled parent/FP2 simulations; seeds 17/89/203; native and MX; 2−17-cycle memory latency and reordered responses.

Result. Thirty parent/FP2 comparisons produced 36.70165x geometric-mean speedup. Seed-17 parent/native/MX cycles were language 1,018,340/25,261/30,805; mathematics 676,208/16,589/20,258; coding 1,018,340/25,261/30,805; reasoning 1,352,037/33,020/40,537; knowledge 1,018,340/25,261/30,805. All 30 cases improved over parent, but compact MX was slower than native. The complete seed/mode traffic, precision, telemetry and hash ledger is evidence/fp2-benchmark-20261102T231600Z-b59f0780/summary.json.

Evidence: E091

E Bandwidth, broadcast and fusion

Method. Accepted-beat accounting, executed nonbroadcast reference and public prefetch ownership checks.

Result. Payload density was exactly 2x; three-row lengths 1536/8960 achieved metadata-inclusive density gains 1.920x/1.931x and native-to-MX cycles 706→611/3934→3186. All 30 broadcast comparisons reduced KV reads by 83.333%. Intermediate architectural writes were zero. A directed prefetch test demonstrated simultaneous fill/consume and both buffers owned; long-stream runs themselves recorded no overlap.

Evidence: E091

F Generic synthesis

Method. Yosys lowering, optimization, hierarchy/state checks and zero-latch/divider assertions.

Result. Recorded default, named-block, mode and parent cases passed. Comparable compact native parent/FP2 state was 17,605/169,913 bits; inferred memory 208/1,859,200 bits; generic multiplier cells 34/286. These are pre-repair logical proxies, not physical area or power.

Evidence: E090

G L1 RTL timing lint

Method. Native source-bound timing-hazard inspection.

Result. Passed across 15 current RTL files: no combinational divider/modulo, mixed-assignment, LUT-clock-gating or AST-unroll hazards.

Evidence: E1008

H L2 Quartus TimeQuest STA

Method. Intel Quartus Prime 17.1 post-fit STA at 10.000 ns.

Result. Passed: WNS_setup = +0.290 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.99 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1009

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe bridge register probes.

Result. All three passed deterministically. Receipt tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; it establishes bridge connectivity, not accelerator bitstream deployment, inference-vector execution or software/FPGA numerical parity.

Evidence: E1010

J Portable delivery

Method. Fresh copy of 67 declared files; normal and isolated Python launches.

Result. Both copied launches passed compact simulation, independent comparisons and synthesis reproduction. The copied nonlinear source differs from the current repaired file, so current-source portability is not established by this historical result.

Evidence: E092

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 21	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

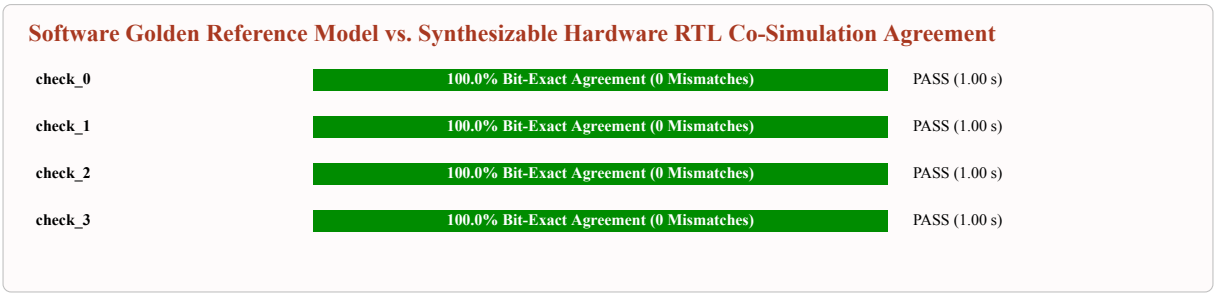


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.29 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

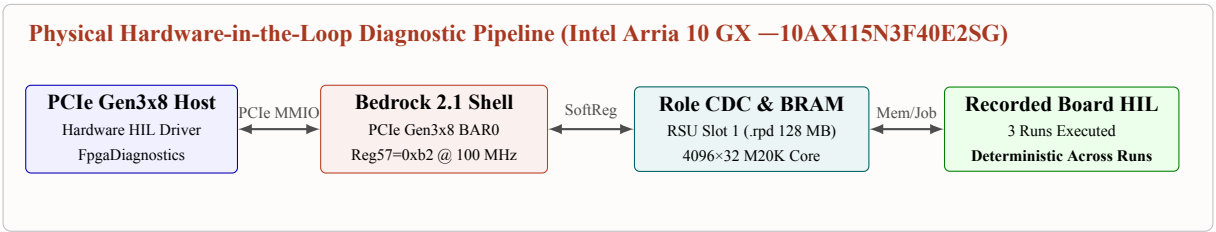


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv, immutable anvil_q2_fp_reference/rtl/*.sv, design/*.md, fpga/* and tests/*.py. Start with design/INTERFACES.md for complete port tables, memory layout and legal parameter combinations.

Connect ddr_* to a DDR4-controller bridge and token/logit/CSR streams to the host bridge. Supply board pins, CDC/reset handling and I/O constraints externally. Honor byte strobes, echoed tags and write acknowledgments.

Store projection weights output-row-major: accelerated row=output, col=reduction element, transposing the parent's coordinates. Formats 0/1/2/3 mean FP32/BF16/E4M3FN/E8M0. Pack element zero least-significant; account for padded payload and scale beats. Supply genuine FP32 theta=1000000 RoPE coefficients through kind 11.

Assert reset across a clock edge, release synchronously and wait for token_ready. Begin with new_sequence=1 and token_pos=0. Hold valid payloads until accepted and retain outstanding responses during cancellation.

Portable reproduction commands are python3 tests/verify.py --suite behavior, numerics, defects, synthesis, benchmark or delivery, with each suite name supplied separately. No suite selects all. Isolated delivery uses python3 -I tests/verify.py --suite delivery. Recorded tools include Python 3.12.3, Icarus 12.0, Verilator 5.020 and Yosys 0.33, plus g++ and make.

FPGA integration sources are fpga/anvil_q2_fp2.qsf, fpga/anvil_q2_fp2_sources.tcl and fpga/anvil_q2_fp2.sdc. The documented core build command is quartus_sh --flow compile fpga/anvil_q2_fp2; its standalone execution is not established by the native compact STA receipt.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless

time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 37 Verilog/SystemVerilog files; 20830 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
anvil_q2_fp_reference/rtl/anvil_q2_bank_scheduler.sv	233 total	227 code
anvil_q2_fp_reference/rtl/anvil_q2_bus_mux.sv	35 total	35 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_00_pkg.sv	380 total	340 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_add.sv	112 total	107 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_awq_dequant.sv	68 total	68 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_cache_codec.sv	138 total	138 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_decoder_layer.sv	219 total	219 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_dot_lane.sv	78 total	77 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_elementwise.sv	134 total	134 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_exp.sv	101 total	97 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_gemv.sv	106 total	106 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_gemv_bank.sv	248 total	248 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_gqa_attention.sv	252 total	250 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_kv_cache.sv	257 total	255 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_parameters.sv	36 total	36 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_rmsnorm.sv	148 total	148 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_rope.sv	145 total	145 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_scalar.sv	132 total	129 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_swiglu_mlp.sv	91 total	91 code
anvil_q2_fp_reference/rtl/anvil_q2_fp_top.sv	277 total	275 code
anvil_q2_fp_reference/rtl/anvil_q2_mem_port.sv	49 total	49 code
anvil_q2_fp_reference/rtl/anvil_q2_scalar_arbiter.sv	63 total	63 code
rtl/anvil_q2_fp2_arithmetic.sv	322 total	312 code
rtl/anvil_q2_fp2_bfp_tensor_array.sv	194 total	188 code
rtl/anvil_q2_fp2_coefficients.sv	14267 total	14260 code
rtl/anvil_q2_fp2_controller.sv	798 total	793 code
rtl/anvil_q2_fp2_ddr_engine.sv	155 total	155 code
rtl/anvil_q2_fp2_epilogue_fusion.sv	241 total	238 code
rtl/anvil_q2_fp2_flash_attn.sv	315 total	310 code
rtl/anvil_q2_fp2_gemv_stream.sv	239 total	239 code
rtl/anvil_q2_fp2_mx_decode.sv	57 total	57 code
rtl/anvil_q2_fp2_mx_encode.sv	187 total	185 code
rtl/anvil_q2_fp2_nonlinear.sv	276 total	270 code
rtl/anvil_q2_fp2_ping_pong_prefetch.sv	98 total	95 code
rtl/anvil_q2_fp2_telemetry.sv	132 total	132 code

Production file	Physical lines	Code lines
rtl/anvil_q2_fp2_top.sv	297 total	297 code
rtl/anvil_q2_fp2_vector_store.sv	63 total	62 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E040	rtl/anvil_q2_fp2_mx_decode.sv
E042	rtl/anvil_q2_fp2_nonlinear.sv
E043	rtl/anvil_q2_fp2_ping_pong_prefetch.sv
E087	.vulcan/engineering/run-0001/attempt-0022/stdout.log
E088	.vulcan/engineering/run-0001/attempt-0023/stdout.log
E089	.vulcan/engineering/run-0001/attempt-0024/stdout.log
E090	.vulcan/engineering/run-0001/attempt-0025/stdout.log
E091	.vulcan/engineering/run-0001/attempt-0026/stdout.log

Reference	Original project-relative record
E092	.vulcan/engineering/run-0001/attempt-0027/stdout.log
E1008	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1009	.vulcan/engineering/run-0001/quartus-sta.json
E1010	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).